

CHATBOT FOR UNIVERSITY INQUIRIES USING RASA NLU

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ABSTRACT

This study develops a university inquiry chatbot for CvSU Imus Campus using Rasa NLU, achieving 93.8% intent classification accuracy across 28 intent categories for enrollment, fees, and schedule queries.

INTRODUCTION

University registrar and administrative offices at CvSU Imus receive hundreds of repetitive inquiries daily regarding enrollment, tuition fees, class schedules, and academic calendar. A conversational AI chatbot can handle routine queries 24/7, reducing staff workload. This study builds and evaluates a domain-specific chatbot using Rasa's open-source NLU framework.

METHODOLOGY

Rasa NLU 3.0 was used with the DIETClassifier pipeline (transformer-based intent and entity extraction). The training dataset contains 1,840 annotated utterances across 28 intents collected from actual student inquiries. Responses are templated for factual queries and use Rasa forms for multi-turn enrollment workflows. The bot was deployed as a Facebook Messenger integration. Evaluation: intent classification accuracy, entity F1, and user satisfaction survey.

RESULTS

The DIETClassifier achieved 93.8% intent classification accuracy on the held-out test set. Entity F1 (for dates, program names, subject codes) was 91.2%. In a 2-week live pilot, the bot handled 1,247 conversations with a 79% resolution rate (without human handoff). Average user satisfaction was 4.1/5. The most common failure mode was out-of-scope queries (14.3% of sessions).

DISCUSSION

Rasa NLU provides a practical and customizable framework for domain-specific university chatbots. The 79% resolution rate is comparable to similar academic chatbot implementations. Out-of-scope handling requires a more robust fallback strategy; adding a catch-all GPT-based response generator is a recommended future enhancement. Full integration with CvSU's SIS for real-time enrollment data remains the highest-priority

improvement.